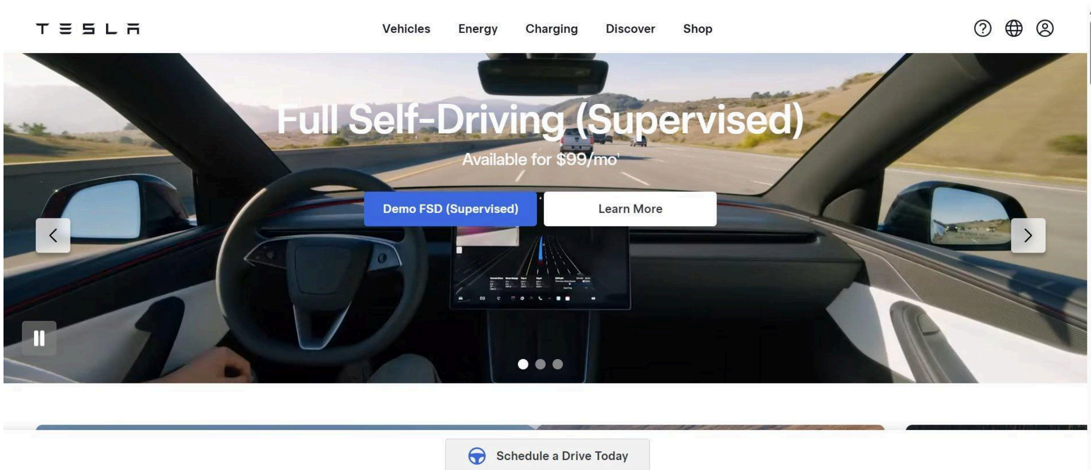
PASSIVE ATTACK SURFACE RECONNAISSANCE REPORT

Target: tesla.com



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Date: March 13, 2026
Type: Passive OSINT | CTI Research Portfolio
Environment: Kali Linux

All activities performed using passive, publicly available data only.
No systems were accessed, scanned, or exploited.

1. Introduction

This report documents a complete passive attack surface reconnaissance exercise performed against tesla.com on March 13, 2026. The purpose of this exercise is to demonstrate real-world Cyber Threat Intelligence (CTI) methodology using exclusively publicly available data and open-source tools — with no active scanning, exploitation, or unauthorized access to any system.

The methodology follows a structured threat intelligence workflow: starting from a single seed domain and progressively expanding the attack surface through multiple intelligence layers including host intelligence, DNS enumeration, SSL certificate transparency logs, ASN discovery, and internet scanning platforms.

1.1 Scope & Legal Disclaimer

All data collected during this exercise was sourced from publicly available databases and services including:

- Host.io — domain intelligence platform
- Certificate Transparency Logs (crt.sh)
- Public WHOIS / RDAP registries (ARIN, MarkMonitor)
- DNS infrastructure (dig, dnsx)
- Shodan — internet-facing device search engine
- BGP routing databases (RADB)
- OSINT subdomain tools (Assetfinder, httprobe)

This report is produced for educational and portfolio purposes only. Tesla, Inc. operates a public bug bounty program through Bugcrowd, and the information collected here reflects only what is already publicly accessible to any internet user.

2. Methodology & Tools

The recon pipeline follows an 8-step passive intelligence methodology:

Step	Phase	Tools Used	Output
1	Host Intelligence	Host.io API	Co-hosted, redirects, backlinks
2	DNS Records Analysis	dig, curl (Host.io DNS API)	A, NS, MX records
3	Reverse NS Lookup	ViewDNS.info	Dead end (Akamai shared NS)
4	Reverse WHOIS	whois, MarkMonitor	Dead end (Privacy proxy)
5	SSL Cert Enumeration	crt.sh certificate transparency	505 subdomains
6	Subdomain OSINT	Assetfinder	+19 unique subdomains
7	Live Detection	httprobe	205 live services
8	IP & ASN Discovery	whois, ARIN RDAP, RADB, Shodan	7 IP blocks, 2 ASNs

2.1 Tool Descriptions

Host.io

A domain intelligence platform that provides co-hosted domains, backlinks, redirects, and DNS data via a REST API. Used in Step 1 to collect initial domain footprint.

crt.sh (Certificate Transparency)

A public database of all SSL/TLS certificates issued by trusted Certificate Authorities. Because certificates must be publicly logged, this is one of the most reliable subdomain discovery methods, bypassing all privacy protections. Queried using wildcard pattern %.tesla.com.

Assetfinder

A Go-based OSINT tool by Tom Hudson that queries multiple public sources (crt.sh, certspotter, hackertarget, Wayback Machine, etc.) to discover subdomains. Used to supplement crt.sh results.

httprobe

A Go-based tool by Tom Hudson that sends concurrent HTTP/HTTPS requests to a list of domains and returns only those that respond with a valid HTTP status. Used to filter the 524 discovered subdomains down to 205 live services.

WHOIS / ARIN RDAP

The Internet Registry database used to identify IP block ownership, ASN assignments, and organizational data. ARIN (American Registry for Internet Numbers) maintains records for North American IP space.

RADB (Routing Assets Database)

A public routing database that stores BGP routing policies. Used to enumerate all IP prefixes announced by a specific Autonomous System Number (ASN).

Shodan

An internet-wide scanning platform that indexes publicly accessible devices, open ports, and service banners. Used passively to count and categorize Tesla-owned IP addresses.

3. Step-by-Step Findings

3.1 Phase 1 — Host Intelligence (Host.io)

The first step in the methodology is to query the Host.io platform for the target domain. This provides three key intelligence sources: co-hosted domains (other domains sharing the same IP), backlinks (domains linking to tesla.com), and redirects (domains that redirect to tesla.com).

Two API calls were made to Host.io — one for web intelligence and one for DNS records:

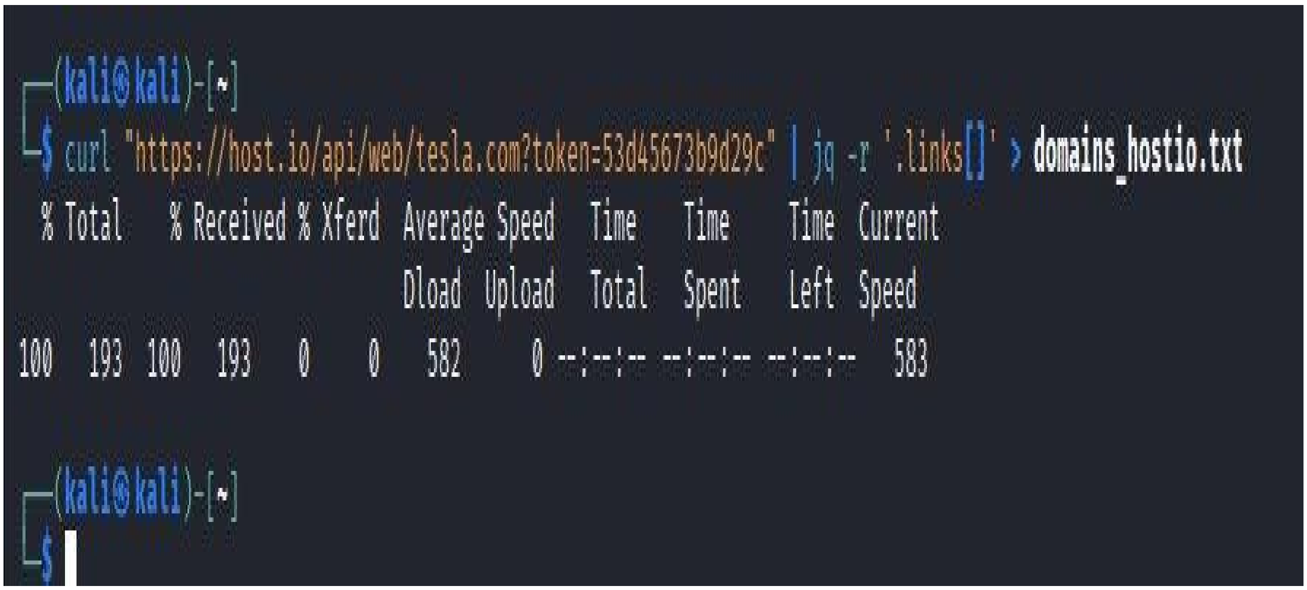


Figure 1 — Host.io web intelligence API call for tesla.com

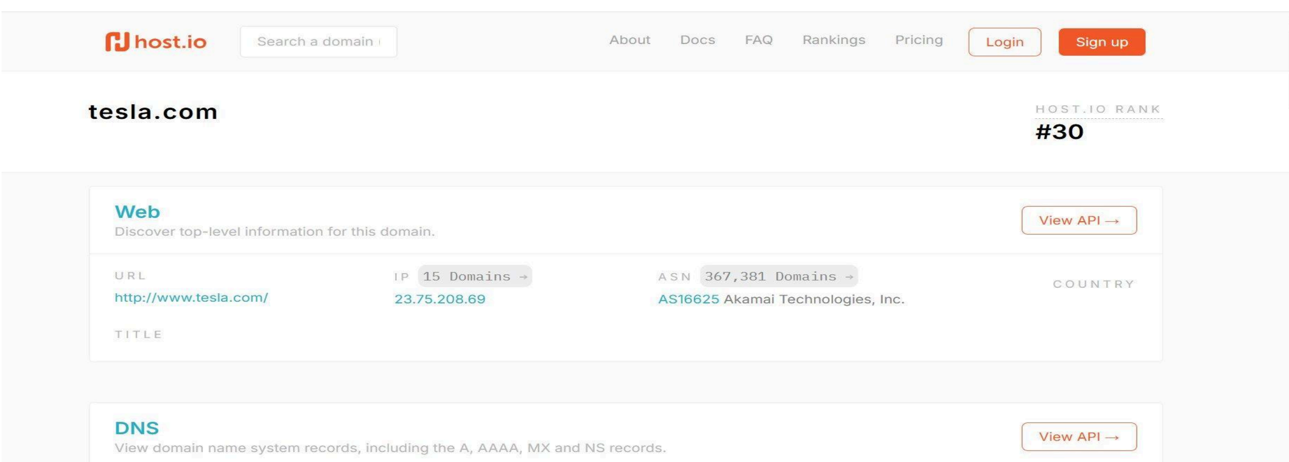


Figure 2 — Host.io dashboard: tesla.com ranked #30 globally, hosted on Akamai AS16625

Co-Hosted Domains

Host.io identified 15 co-hosted domains on the same Akamai IP (23.75.208.69). Most are unrelated third-party sites sharing the same CDN infrastructure — confirming that co-hosted data from a shared CDN like Akamai carries low intelligence value for identifying Tesla-specific assets.

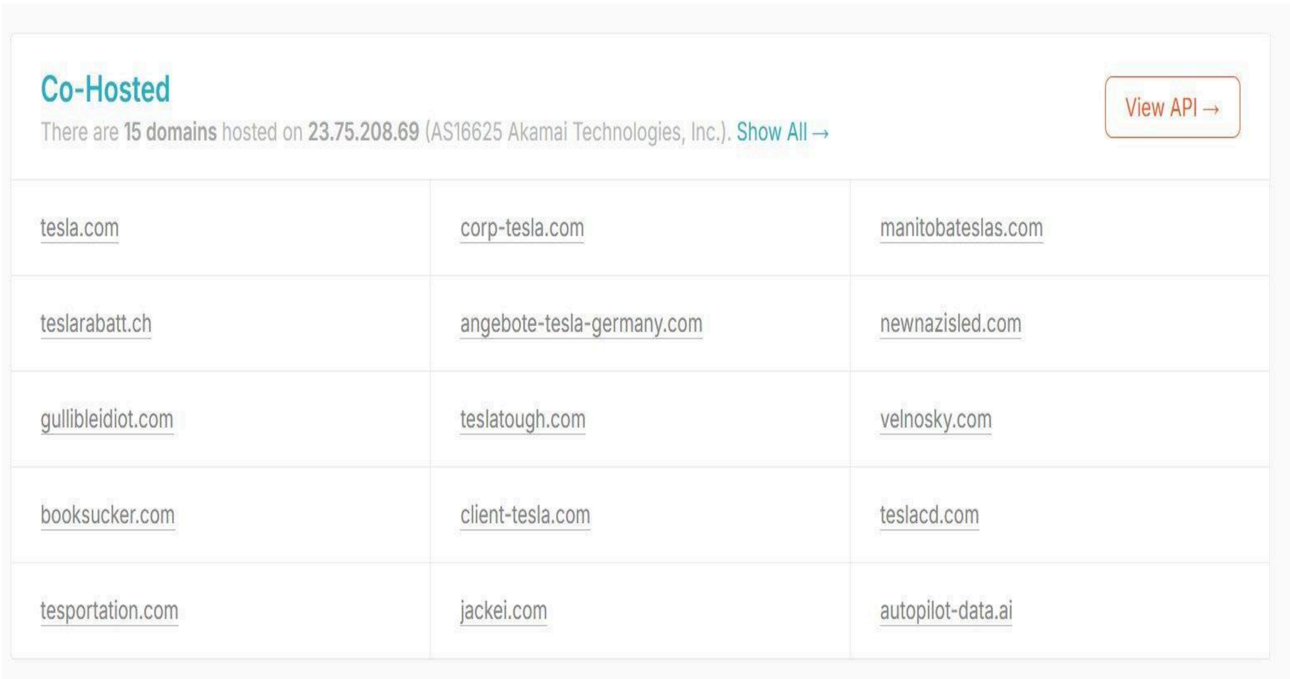


Figure 3 — Co-hosted domains on the same Akamai IP as tesla.com

Backlinks

Host.io reported 2,989 domains backlinking to tesla.com. These are primarily third-party sites, news outlets, partner sites, and community resources. Notable backlinks include karpathy.ai (former Tesla AI Director Andrej Karpathy's personal site) and supercharge.info, confirming legitimate association.

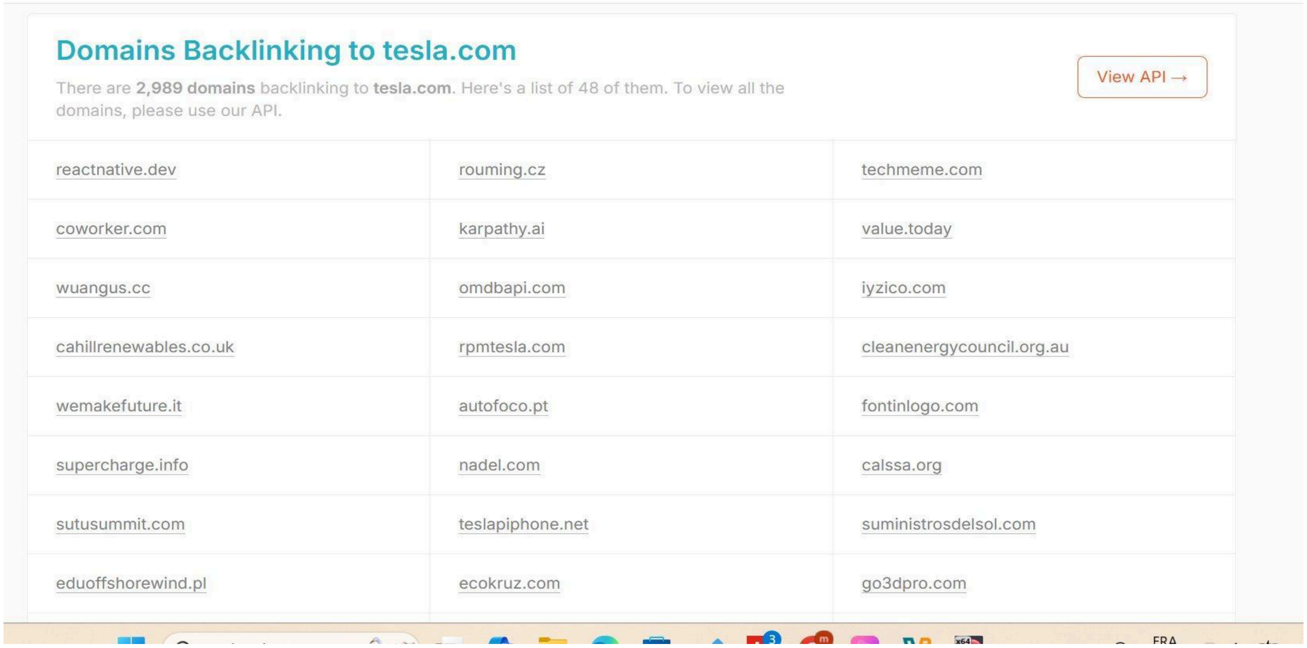


Figure 4 — 2,989 domains backlinking to tesla.com (sample shown)

Redirects

Host.io identified 1,053 domains redirecting to tesla.com. These include branded regional domains, typosquatted domains, and legacy domains Tesla has acquired or controls. Notable entries include ts.la (Tesla's official short link), teslaenergy.nu, and teslamotor-benelux.com.

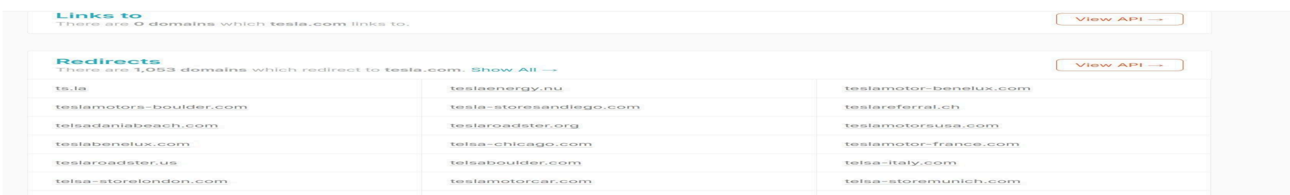


Figure 5 — Sample of 1,053 redirect domains (ts.la, teslaenergy.nu, regional domains)

DNS Records via Host.io API

The Host.io DNS API endpoint was used to retrieve A, NS, and MX records programmatically:

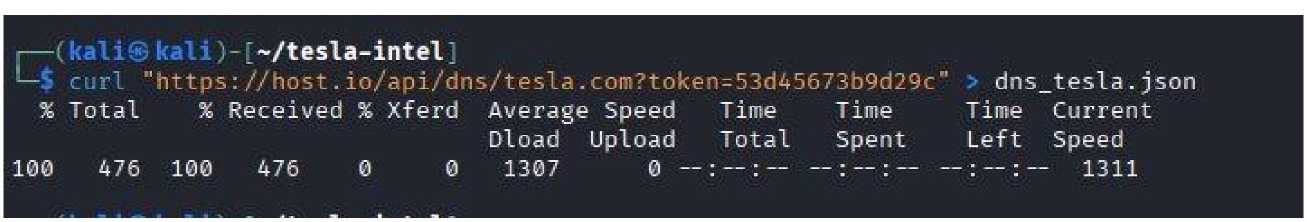


Figure 6 — Host.io DNS API call saving results to dns_tesla.json

3.2 Phase 2 — DNS Records Analysis

The DNS records were parsed from the JSON output to extract A records (IP addresses), NS records (nameservers), and MX records (mail servers).

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat dns_tesla.json
{
  "domain": "tesla.com",
  "a": [
    "2.18.48.207",
    "2.18.49.207",
    "2.18.50.207",
    "2.18.51.207",
    "2.18.52.207",
    "2.18.53.207",
    "2.18.54.207",
    "2.18.55.207",
    "23.40.100.207",
    "23.7.244.207"
  ],
  "mx": [
    "tesla-com.mail.protection.outlook.com."
  ],
  "ns": [
    "a1-12.akam.net.",
    "a10-67.akam.net.",
    "a12-64.akam.net.",
    "a28-65.akam.net.",
    "a7-66.akam.net.",
    "a9-67.akam.net.",
    "edns69.ultradns.com."
  ]
}
```

Figure 7 — dns_tesla.json: A records resolving to Akamai IPs, MX to Office 365, NS to Akamai/UltraDNS

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat dns_tesla.json | jq -r '.a[]' > tesla_ips.txt

┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat dns_tesla.json | jq -r '.ns[]' > tesla_nameservers.txt

┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat dns_tesla.json | jq -r '.mx[]' > tesla_mx.txt

┌──(kali㉿kali)-[~/tesla-intel]
```

Figure 8 — Extracting A records, nameservers and MX records via jq

Key DNS findings:

- A Records: Multiple Akamai IP addresses (2.18.48-55.207, 23.40.100.207, 23.7.244.207) — confirming CDN-fronted infrastructure
- MX Record: tesla-com.mail.protection.outlook.com — Tesla uses Microsoft Office 365 for email
- NS Records: 6x Akamai nameservers (akam.net) + UltraDNS (edns69.ultradns.com) — dual-DNS redundancy

3.3 Phase 3 — Reverse NS Lookup

A Reverse NS Lookup was performed on the discovered nameservers using ViewDNS.info to attempt to find other domains registered under the same nameserver — potentially revealing additional Tesla-owned domains.

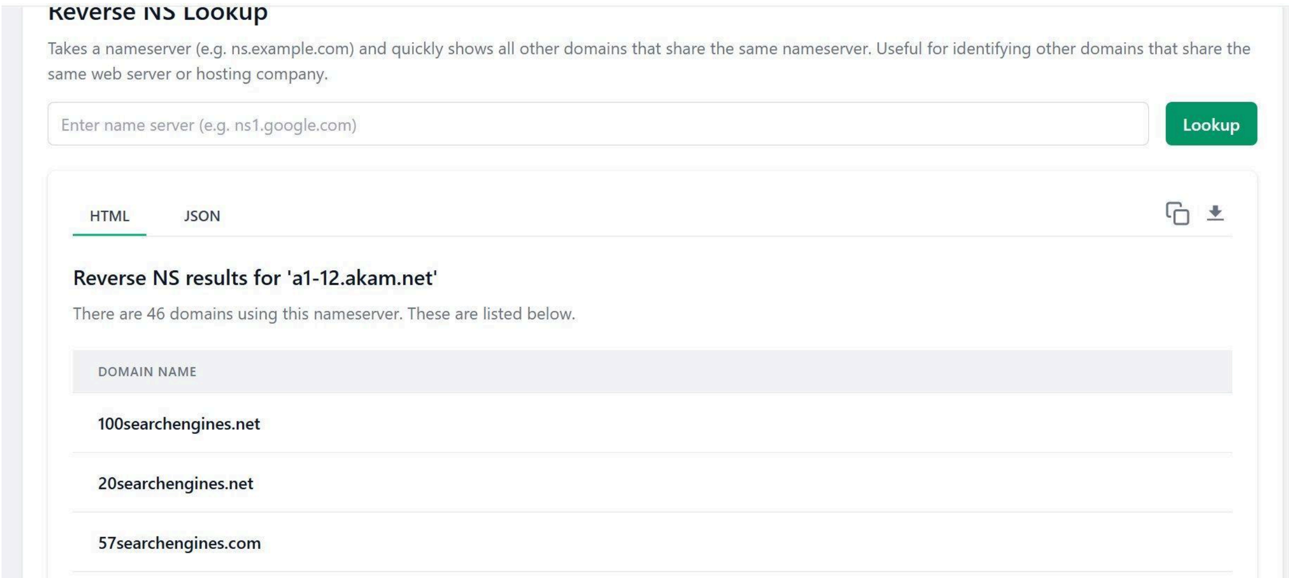


Figure 9 — ViewDNS.info Reverse NS Lookup for a1-12.akam.net (46 results, none Tesla-related)

Result: Dead end. Tesla uses Akamai — one of the world's largest CDNs with hundreds of thousands of customers sharing the same nameservers. The reverse NS lookup returned 46 completely unrelated domains. The same pattern applied to all other Akamai nameservers (a10-67.akam.net returned 550+ unrelated domains). Reverse NS was therefore skipped as an intelligence source for this target.

3.4 Phase 4 — Reverse WHOIS

WHOIS data was queried directly from the MarkMonitor registrar to attempt identification of the registrant organization or email — which could then be used for Reverse WHOIS to discover other Tesla-registered domains.

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ whois -h whois.markmonitor.com tesla.com
Domain Name: tesla.com
Registry Domain ID: 187902_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-10-02T10:15:20+0000
Creation Date: 1992-11-04T05:00:00+0000
Registrar Registration Expiration Date: 2026-11-03T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Name: Domain Administrator
Registrant Organization: DNStination Inc.
Registrant Street: 3450 Sacramento Street, Suite 405
Registrant City: San Francisco
Registrant State/Province: CA
Registrant Postal Code: 94118
Registrant Country: US
Registrant Phone: +1.4155319335
Registrant Phone Ext:
Registrant Fax: +1.4155319336
Registrant Fax Ext:
Registrant Email: admin@dnstinations.com
Tech Name: Domain Administrator
Tech Phone: +1.4155319335
Tech Email: admin@dnstinations.com
Name Server: a1-12.akam.net
Name Server: a10-67.akam.net
Name Server: a12-64.akam.net
Name Server: a28-65.akam.net
Name Server: edns69.ultradns.net
Name Server: a9-67.akam.net
Name Server: edns69.ultradns.biz
Name Server: edns69.ultradns.org
Name Server: edns69.ultradns.com
Name Server: a7-66.akam.net
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-03-13T12:20:13+0000 <<<
```

Figure 10 — WHOIS query on tesla.com: registrant hidden behind DNStination Inc. privacy proxy

Result: Dead end. Tesla uses DNStination Inc. (San Francisco, CA) as a domain privacy proxy service — essentially a P.O. box for domain registrations. The registrant email (admin@dnstinations.com) and organization name are not Tesla's real details. Reverse WHOIS on this proxy would return thousands of unrelated domains. This step was skipped in favor of more productive methods.

Additional notable findings from the WHOIS record:

- Domain created: November 4, 1992 — one of the oldest .com domains
- Registrar: MarkMonitor (used exclusively by large enterprises for domain protection)
- All 6 ICANN EPP locks active: clientUpdate, clientTransfer, clientDelete, serverUpdate, serverTransfer, serverDelete — maximum domain protection

3.5 Phase 5 — SSL Certificate Enumeration (crt.sh)

SSL Certificate Transparency logs represent one of the most powerful passive subdomain discovery techniques. Every SSL/TLS certificate issued by a trusted CA must be publicly logged in Certificate Transparency logs. The crt.sh platform aggregates these logs and provides a searchable API.

A wildcard query (%.tesla.com) was issued against crt.sh:

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ curl -s "https://crt.sh/?q=%25.tesla.com&output=json" \
  | python3 -c "
import json,sys
data=json.load(sys.stdin)
for d in data:
    for name in d['name_value'].split('\n'):
        print(name.strip())
" | sort -u > ~/tesla-intel/crt_subdomains.txt

wc -l ~/tesla-intel/crt_subdomains.txt
cat ~/tesla-intel/crt_subdomains.txt | head -20
505 /home/kali/tesla-intel/crt_subdomains.txt
acme-sentry-4a.eng.use1.vn.cloud.tesla.com
acme-sentry-4.eng.use1.vn.cloud.tesla.com
acs2-poc.voice.tesla.com
ai-api-stg.tesla.com
ai-api.tesla.com
ai-api-uat.tesla.com
akamai-apigateway-einvoicing-stg.tesla.com
akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
akamai-apigateway-vehicleextinfogw-stgsvc-st.tesla.com
ams13-gpgw1.tesla.com
apac-cppm.tesla.com
*.apac.logs.tesla.com
apacvpn1.tesla.com
apacvpn2.tesla.com
apacvpn.tesla.com
apf-api.eng.vn.cloud.tesla.com
apf-api.prd.vn.cloud.tesla.com
api-account-master.tesla.com
api-firebolt.tesla.com
api.kronos.tesla.com
```

Figure 11 — crt.sh query returning 505 unique subdomains for tesla.com

This single query produced 505 subdomains — making crt.sh the single most productive intelligence source in this exercise, with no API key or authentication required.

Wildcard Certificate Entries

28 wildcard certificate entries were identified, each revealing an entire subdomain namespace:

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ # Check for wildcard entries (*.something) - note them separately
grep "^\*\." ~/tesla-intel/crt_subdomains.txt > ~/tesla-intel/wildcards.txt
echo "Wildcards found:"
cat ~/tesla-intel/wildcards.txt

# Clean list - remove wildcards from main list
grep -v "^\*\." ~/tesla-intel/crt_subdomains.txt > ~/tesla-intel/clean_subdomains.txt
wc -l ~/tesla-intel/clean_subdomains.txt
Wildcards found:
*.apac.logs.tesla.com
*.app.tesla.com
*.bottlerocket.tesla.com
*.ca.tesla.com
*.cloud.tesla.com
*.cn.tesla.com
*.de.tesla.com
*.energy.tesla.com
*.engage.tesla.com
*.engagetesla.com
*.eu.logs.tesla.com
*.factory-berlin.tesla.com
*.gf.tesla.com
*.github.tesla.com
*.logs.tesla.com
*.manager.courses.tesla.com
*.mfg.tesla.com
*.na.logs.tesla.com
*.nv.tesla.com
*.ny.tesla.com
*.obs.tesla.com
*.paloalto.tesla.com
*.powerhub.energy.tesla.com
*.sandbox-manager.courses.tesla.com
*.tesla.com
*.tx.tesla.com
*.vdi.tesla.com
*.zpra.tesla.com
477 /home/kali/tesla-intel/clean_subdomains.txt
```

*Figure 12 — 28 wildcard entries including *.factory-berlin.tesla.com, *.gf.tesla.com, *.mfg.tesla.com, *.vdi.tesla.com*

Notable wildcard domains and their implications:

- *.factory-berlin.tesla.com — Gigafactory Berlin dedicated namespace
- *.gf.tesla.com — Gigafactory infrastructure
- *.mfg.tesla.com — Manufacturing systems
- *.vdi.tesla.com — Virtual Desktop Infrastructure (remote worker access)
- *.cn.tesla.com — China operations
- *.bottlerocket.tesla.com — AWS Bottlerocket container OS infrastructure
- *.engagetesla.com — A completely separate Tesla domain for engagement campaigns

VPN Endpoints

SSL certificates revealed Tesla's VPN infrastructure, including regional endpoints:

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ grep "vpn" ~/tesla-intel/crt_subdomains.txt
apacvpn1.tesla.com
apacvpn2.tesla.com
apacvpn.tesla.com
cnvpn1.tesla.com
cnvpn.tesla.com
vpn1.tesla.com
vpn2.tesla.com
vpn3.tesla.com
vpn.tesla.com
www.vpn1.tesla.com

┌──(kali㉿kali)-[~/tesla-intel]
└─$ grep -E "\.eng\.|staging|stg|dev|uat|poc|internal" ~/tesla-intel/crt_subdomains.txt
acme-sentry-4a.eng.use1.vn.cloud.tesla.com
acme-sentry-4.eng.use1.vn.cloud.tesla.com
acs2-poc.voice.tesla.com
ai-api-stg.tesla.com
ai-api-uat.tesla.com
akamai-apigateway-einvoicing-stg.tesla.com
akamai-apigateway-vehicleextinfogw-stgsvc-st.tesla.com
apf-api.eng.vn.cloud.tesla.com
```

Figure 13 — VPN endpoints and staging/dev environments found in certificate logs

- vpn.tesla.com, vpn1/2/3.tesla.com — main VPN infrastructure
- apacvpn.tesla.com, apacvpn1/2.tesla.com — Asia-Pacific VPN
- cnvpn.tesla.com, cnvpn1.tesla.com — China VPN endpoints

Internal & Staging Environments

A significant number of staging, development, and POC environments were discovered through certificate logs:

- sso-dev.tesla.com — SSO development environment (publicly DNS-resolvable)
- kronos-dev.tesla.com — Workforce management dev environment
- mfa-dev, mfamobile-dev, mfauser-dev.tesla.com — MFA systems
- auth-stage.tesla.com — Authentication staging
- www-stg2, www-uat2, www-uat.tesla.com — Website staging environments
- tf-poc.tesla.com — Proof of concept environment

API Endpoints

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ grep "api" ~/tesla-intel/crt_subdomains.txt
ai-api-stg.tesla.com
ai-api.tesla.com
ai-api-uat.tesla.com
akamai-apigateway-einvoicing-stg.tesla.com
akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
akamai-apigateway-vehicleextinfogw-stgsvc-st.tesla.com
apf-api.eng.vn.cloud.tesla.com
```

Figure 14 — API subdomains discovered: ai-api, fleet-api, ownerapi, tvs-api, media-server-api

Cloud Infrastructure

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ grep "cloud" ~/tesla-intel/crt_subdomains.txt
acme-sentry-4a.eng.use1.vn.cloud.tesla.com
acme-sentry-4.eng.use1.vn.cloud.tesla.com
apf-api.eng.vn.cloud.tesla.com
apf-api.prd.vn.cloud.tesla.com
auth.eng.euw1.vn.cloud.tesla.com
auth.eng.usw.vn.cloud.tesla.com
```

Figure 15 — Cloud infrastructure subdomains: vn.cloud.tesla.com namespace (vehicle network)

The vn.cloud.tesla.com namespace appears to be Tesla's Vehicle Network cloud infrastructure — handling vehicle telemetry, owner API, fleet management, media server, and signaling across multiple global regions (usw, usw2, euw1, na).

Merging Results

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat ~/tesla-intel/crt_subdomains.txt ~/tesla-intel/domains.txt \
  | sort -u > ~/tesla-intel/all_subdomains.txt

wc -l ~/tesla-intel/all_subdomains.txt
505 /home/kali/tesla-intel/all_subdomains.txt
```

Figure 16 — Merging crt.sh results with domains.txt: 505 unique subdomains

3.6 Phase 6 — Subdomain Discovery with Assetfinder

Assetfinder was used to supplement the crt.sh results by querying additional OSINT sources including Wayback Machine, certspotter, hackertarget, and others.

Figure 17 — Assetfinder discovering 53 subdomains for tesla.com

Figure 18 — Merging Assetfinder results with all_subdomains.txt

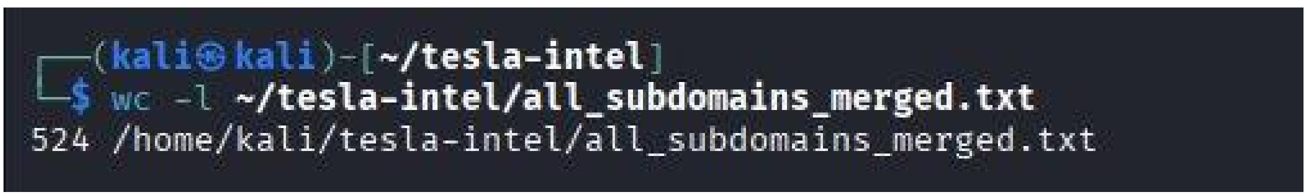


Figure 19 — Final merged list: 524 unique subdomains after deduplication

Assetfinder added 19 unique subdomains not previously found by crt.sh, bringing the total discovered subdomains to 524.

3.7 Phase 7 — Live Subdomain Detection (httprobe)

Having discovered 524 subdomains, the next step was to determine which ones are actually live and responding to HTTP/HTTPS requests. Many certificate entries represent decommissioned services or expired infrastructure. httprobe was used to filter the list.

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat ~/tesla-intel/all_subdomains_merged.txt \
  | httprobe > ~/tesla-intel/alive_subdomains.txt

┌──(kali㉿kali)-[~/tesla-intel]
└─$ wc -l ~/tesla-intel/alive_subdomains.txt
205 /home/kali/tesla-intel/alive_subdomains.txt
```

Figure 20 — httprobe run: 205 live subdomains identified out of 524 (39% live rate)

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ cat ~/tesla-intel/alive_subdomains.txt | head -20

http://akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
http://accounts.tesla.com
https://akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
http://apf-api.eng.vn.cloud.tesla.com
http://apf-api.prd.vn.cloud.tesla.com
http://api-firebolt.tesla.com
https://apf-api.prd.vn.cloud.tesla.com
https://api-firebolt.tesla.com
http://assets-ir.tesla.com
https://accounts.tesla.com
https://apf-api.eng.vn.cloud.tesla.com
https://assets-ir.tesla.com
https://auth.eng.usw.vn.cloud.tesla.com
https://auth.prd.usw.vn.cloud.tesla.com
https://auth.eng.euw1.vn.cloud.tesla.com
https://auth-stage.tesla.com
https://auth.tesla.com
http://auth-stage.tesla.com
https://autobidder.powerhub.energy.tesla.com
http://autobidder-preprd.powerhub.energy.tesla.com
```

Figure 21 — Sample of live subdomains including accounts, auth, autobidder, billing endpoints

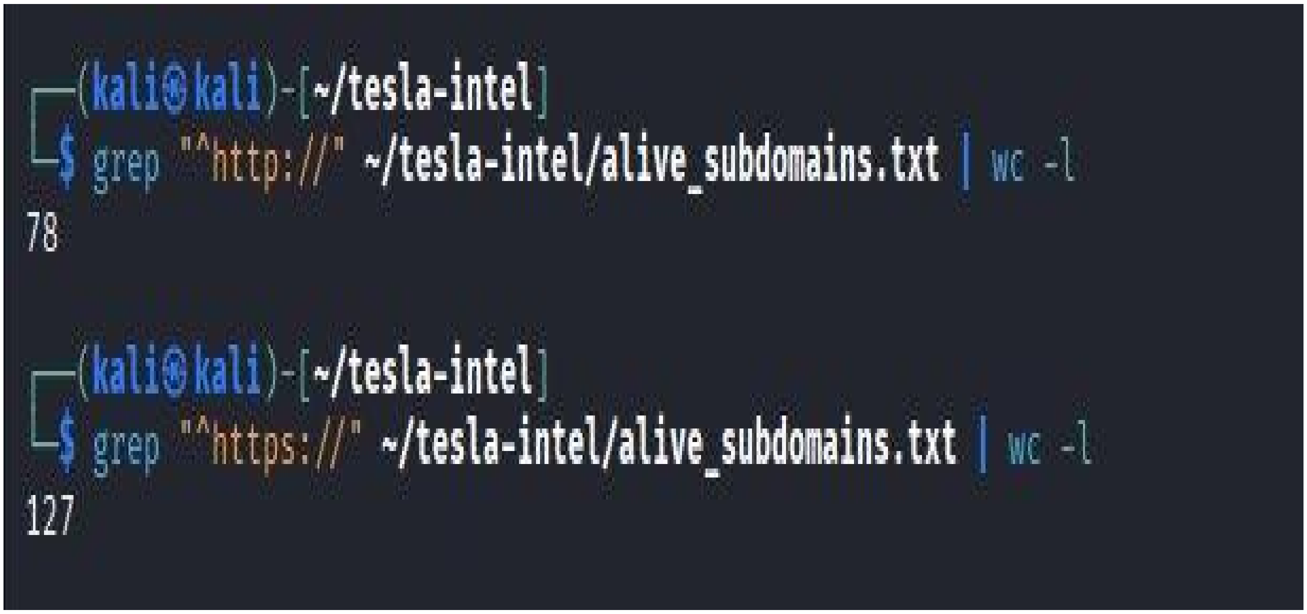
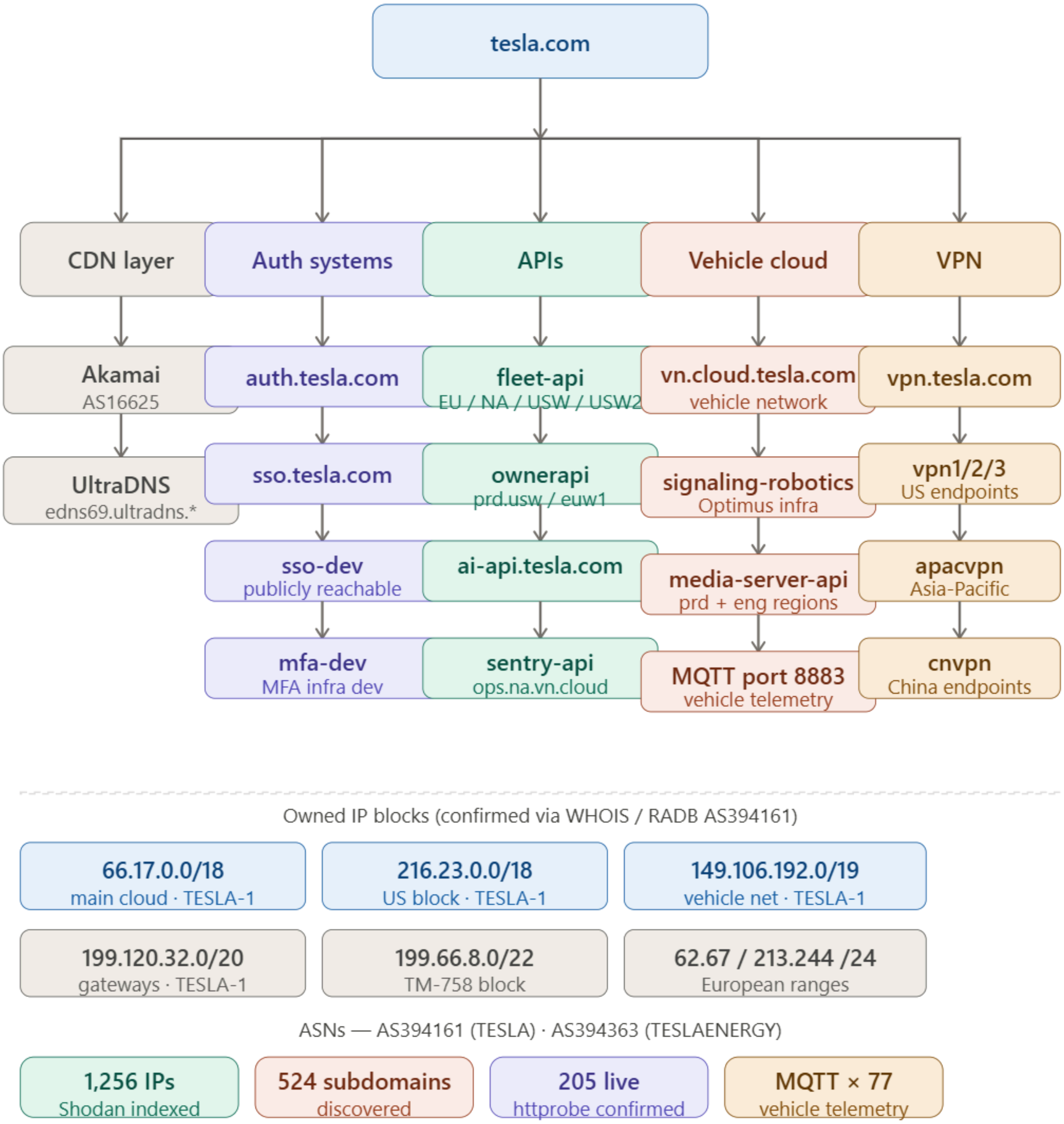


Figure 22 — Protocol breakdown: 78 HTTP responses, 127 HTTPS responses

The 78 HTTP responses were verified to be standard 301 redirects to HTTPS — confirming Tesla enforces HTTPS across all public-facing services. This was spot-checked on sso-dev.tesla.com (301 redirect) and kronos-dev.tesla.com (403 Forbidden on HTTP — indicating IP-restricted access without redirect).

Notable Live Services Discovered


tesla.com
CDN layer
Auth systems
APIs
Vehicle cloud
VPN
Akamai
AS16625
UltraDNS
edns69.ultradns.*
auth.tesla.com
sso.tesla.com
sso-dev
publicly reachable
mfa-dev
MFA infra dev
fleet-api
EU / NA / USW / USW2
ownerapi
prd.usw / euw1
ai-api.tesla.com
sentry-api
ops.na.vn.cloud
vn.cloud.tesla.com
vehicle network
signaling-robotics
Optimus infra
media-server-api
prd + eng regions
MQTT port 8883
vehicle telemetry
vpn.tesla.com
vpn1/2/3
US endpoints
apacvpn
Asia-Pacific
cnvpn
China endpoints
Owned IP blocks (confirmed via WHOIS / RADB AS394161)
66.17.0.0/18
main cloud · TESLA-1
216.23.0.0/18
US block · TESLA-1
149.106.192.0/19
vehicle net · TESLA-1
199.120.32.0/20
gateways · TESLA-1
199.66.8.0/22
TM-758 block
62.67 / 213.244 /24
European ranges
ASNs — AS394161 (TESLA) · AS394363 (TESLAENERGY)
1,256 IPs
Shodan indexed
524 subdomains
discovered
205 live
httprobe confirmed
MQTT × 77
vehicle telemetry

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ grep -E "vpn|admin|sso|mfa|auth|login|dev|stg|api" ~/tesla-intel/alive_subdomains.txt
http://akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
https://akamai-apigateway-vehicleextinfogw-prdsvc-st.tesla.com
http://apf-api.eng.vn.cloud.tesla.com
http://apf-api.prd.vn.cloud.tesla.com
http://api-firebolt.tesla.com
https://apf-api.prd.vn.cloud.tesla.com
https://api-firebolt.tesla.com
https://apf-api.eng.vn.cloud.tesla.com
https://auth.eng.usw.vn.cloud.tesla.com
https://auth.prd.usw.vn.cloud.tesla.com
https://auth.eng.euw1.vn.cloud.tesla.com
https://auth-stage.tesla.com
https://auth.tesla.com
http://auth-stage.tesla.com
http://auth.tesla.com
http://cx-api-apac.tesla.com
http://developer.tesla.com
https://cx-api-apac.tesla.com
https://developer.tesla.com
https://fleet-api.prd.eu.vn.cloud.tesla.com
https://fleet-api.eng.na.vn.cloud.tesla.com
https://fleet-api.prd.na.vn.cloud.tesla.com
https://fleet-api.prd.usw2.vn.cloud.tesla.com
https://fleet-api.prd.usw.vn.cloud.tesla.com
https://fleet-auth.prd.vn.cloud.tesla.com
http://kronos-dev.tesla.com
https://kronos-dev.tesla.com
http://fleet-auth.prd.vn.cloud.tesla.com
https://media-server-api.eng.vn.cloud.tesla.com
https://media-server-api.prd.vn.cloud.tesla.com
https://media-server-dev-api.eng.vn.cloud.tesla.com
https://media-server-qr-dev.eng.america.vn.cloud.tesla.com
https://ownerapi-api.eng.euw1.vn.cloud.tesla.com
https://ownerapi-alpha.prd.vn.cloud.tesla.com
https://ownerapi-api.eng.usw.vn.cloud.tesla.com
https://ownerapi-api.prd.usw.vn.cloud.tesla.com
https://sentry-api.ops.na.vn.cloud.tesla.com
http://sso-dev.tesla.com
http://sso.tesla.com
https://sso-dev.tesla.com
https://sso.tesla.com
```

Figure 23 — Interesting live services: SSO, MFA, auth, fleet-api, ownerapi, sentry-api, kronos-dev

Key live services of note:

- auth.tesla.com / auth-stage.tesla.com — Main authentication infrastructure
- sso.tesla.com / sso-dev.tesla.com — Single Sign-On systems (dev exposed publicly)
- ownerapi-api.prd.usw.vn.cloud.tesla.com — Vehicle owner API (production)
- fleet-api.prd.eu/na/usw/usw2 — Fleet management APIs across 4 global regions
- sentry-api.ops.na.vn.cloud.tesla.com — Sentry monitoring API
- kronos-dev.tesla.com — Workforce management system dev (403 — IP restricted)
- autobidder.powerhub.energy.tesla.com — Tesla Autobidder (energy trading platform)
- signaling-robotics.eng.vn.cloud.tesla.com — Robotics signaling (Optimus robot infra?)

3.8 Phase 8 — IP Address Resolution

All 205 live subdomains were resolved to their IP addresses using dig, producing a mapping of domain to IP. This is the prerequisite step for identifying Tesla's owned IP blocks.

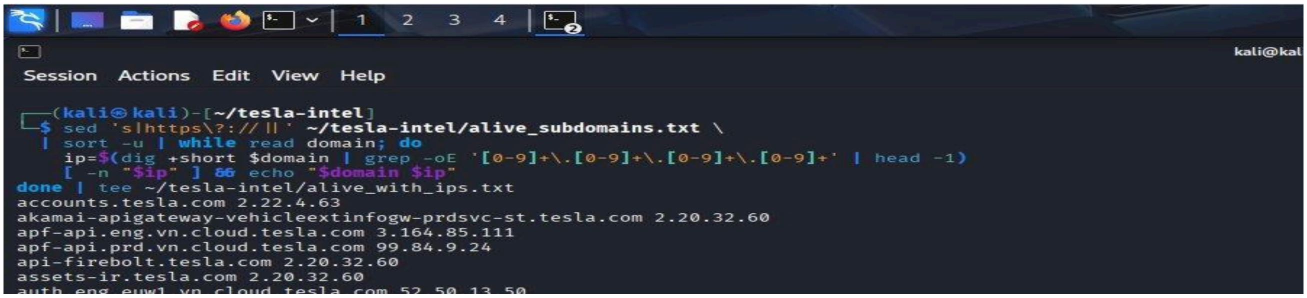


Figure 24 — IP resolution: 130 domains resolved, 70 unique IP addresses identified

Key IP ranges identified from the resolved data:

- 2.20.32.x / 2.22.4.x — Akamai CDN (public-facing tesla.com services)
- 66.17.x.x — Tesla-owned cloud infrastructure (vehicle network, APIs)
- 199.120.x.x — Tesla-owned gateway infrastructure (gpgw1 nodes)
- 18.x / 44.x / 52.x / 54.x — AWS (fleet view, fleet API, authentication)
- 20.x — Microsoft Azure (teslacmg* servers — likely SCCM/MDM)
- 23.x / 151.x / 199.232.x — Fastly CDN (digital assets)
- 184.85.x.x — Akamai (Tesla Energy/Powerhub services)

3.9 Phase 9 — ASN & IP Block Discovery

With IP addresses in hand, the next step was to identify Tesla's owned IP blocks and Autonomous System Numbers using WHOIS, ARIN RDAP, and the RADB routing database.

Initial IP Block Verification

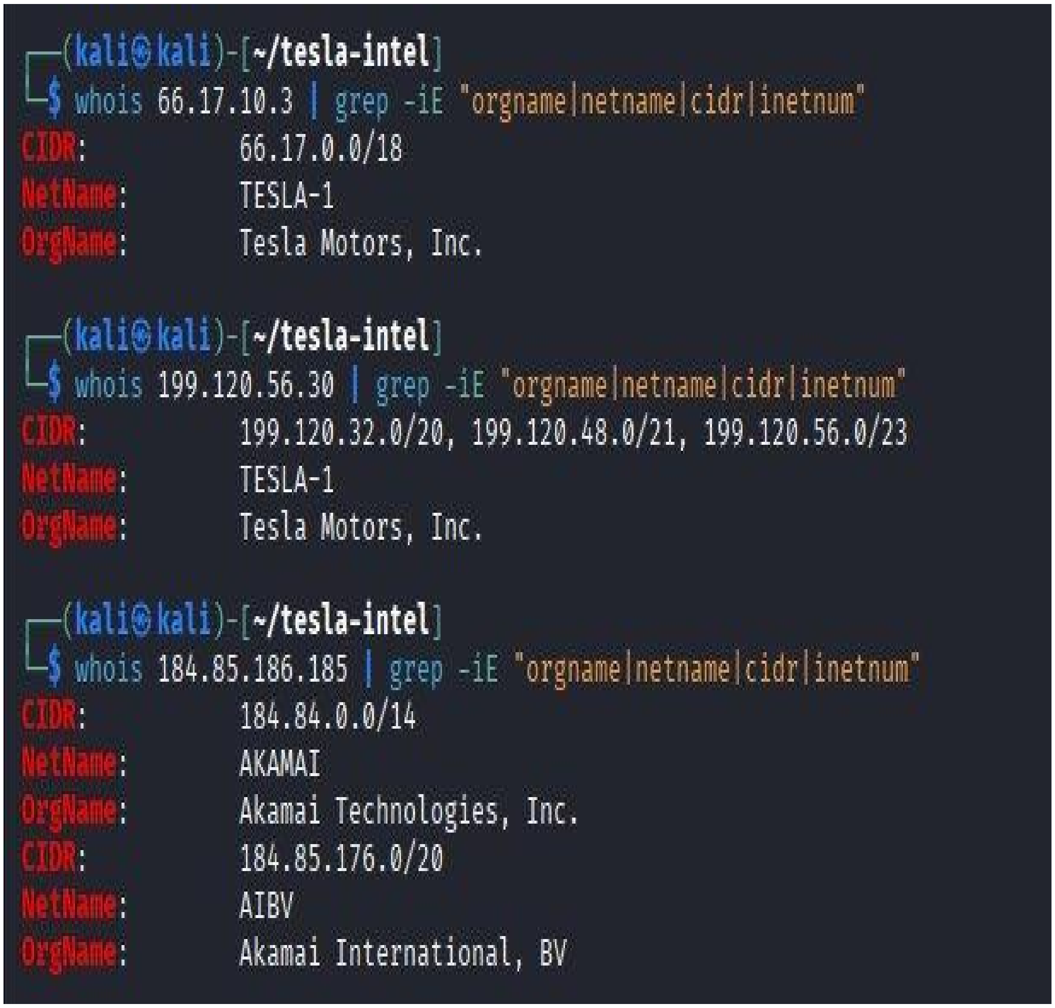


Figure 25 — WHOIS confirmation: 66.17.0.0/18 and 199.120.x/20-23 are TESLA-1 (Tesla Motors, Inc.)

ARIN Organization Handles

Tesla has multiple ARIN organization handles. A systematic lookup of handles TESLA-1 through TESLA-5 revealed two ASNs:

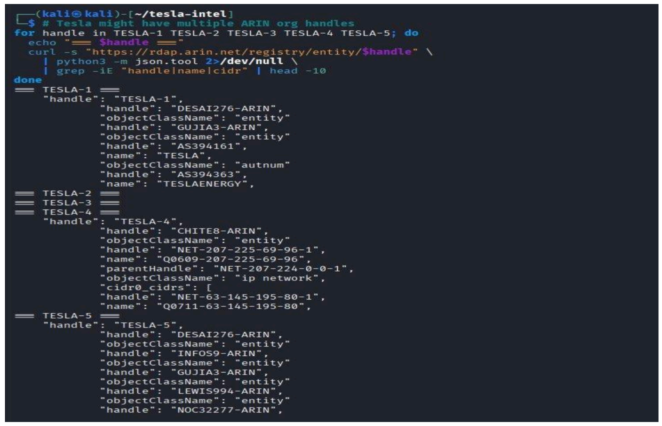


Figure 26 — ARIN RDAP lookup for TESLA-1: reveals AS394161 (TESLA) and AS394363 (TESLAENERGY)

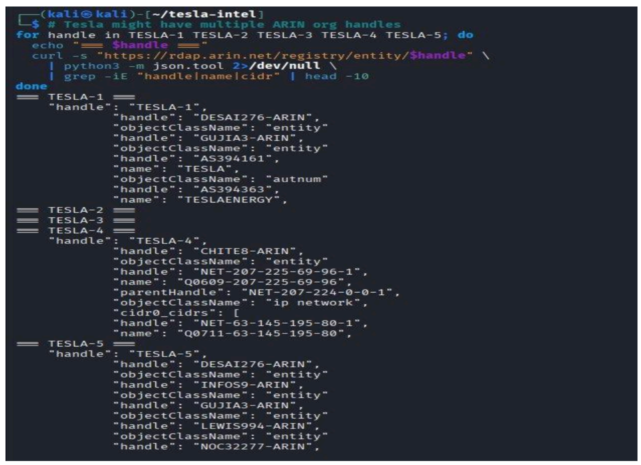


Figure 27 — TESLA-4 handle containing two additional small IP blocks (CenturyLink reassigned)

TESLA-4: Additional IP Blocks

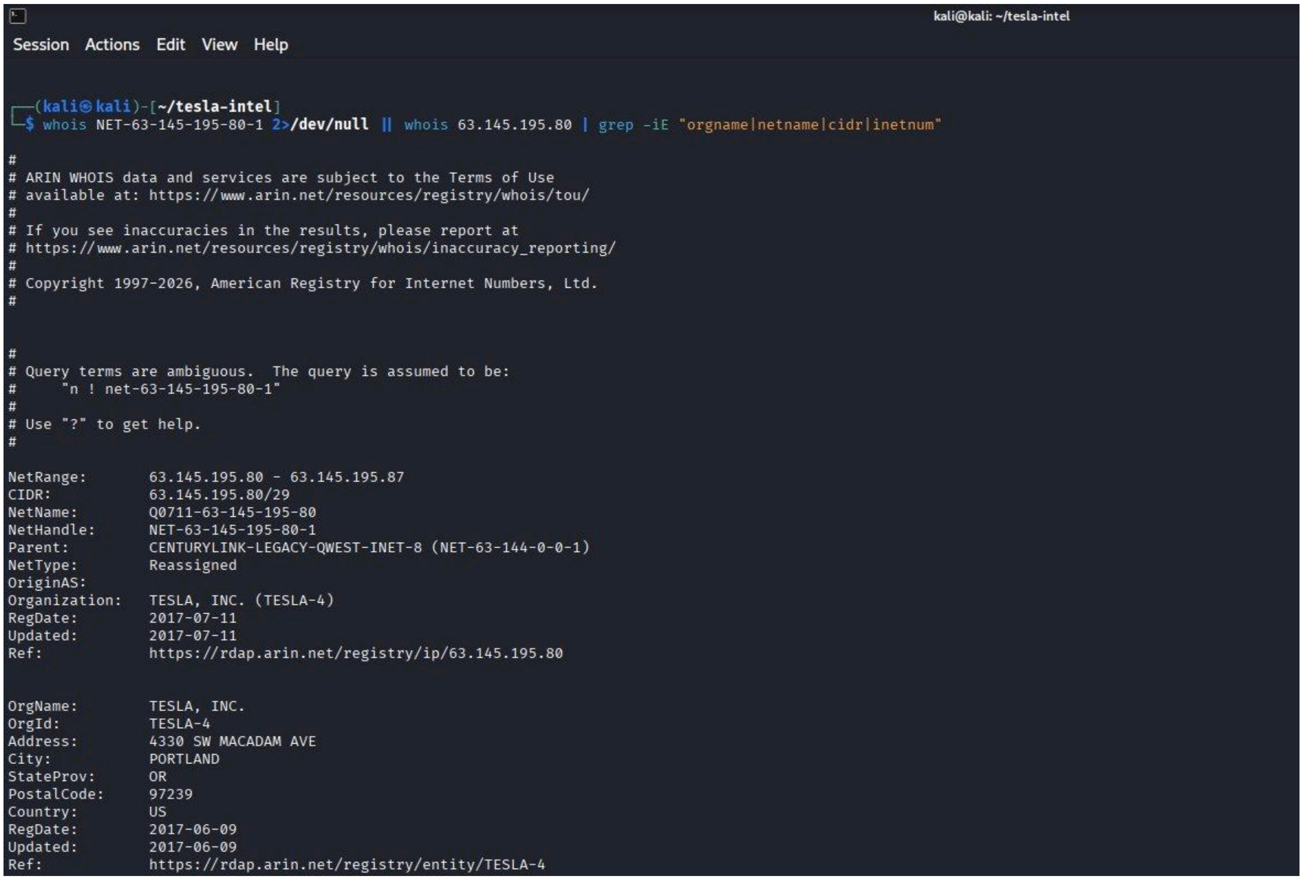
TESLA-4 (registered as TESLA, INC. at 4330 SW Macadam Ave, Portland, OR — a Tesla facility) contains two small /29 blocks reassigned from CenturyLink/Qwest:

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ curl -s "https://rdap.arin.net/registry/entity/TESLA-4" \
  | python3 -m json.tool 2>/dev/null | grep -i "handle\|name"
    "handle": "TESLA-4",
            "handle": "CHITE8-ARIN",
            "objectClassName": "entity"
            "handle": "NET-207-225-69-96-1",
            "name": "Q0609-207-225-69-96",
            "parentHandle": "NET-207-224-0-0-1",
            "objectClassName": "ip network",
            "handle": "NET-63-145-195-80-1",
            "name": "Q0711-63-145-195-80",
            "parentHandle": "NET-63-144-0-0-1",
            "objectClassName": "ip network",
    "objectClassName": "entity"
```

Figure 28 — RDAP entity TESLA-4 showing NET-207-225-69-96-1 and NET-63-145-195-80-1

```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ whois NET-207-225-69-96-1 2>/dev/null || whois 207.225.69.96 | grep -iE "orgname|netname|cidr|inetnum"

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#


#
# Query terms are ambiguous.  The query is assumed to be:
#     "n ! net-207-225-69-96-1"
#
# Use "?" to get help.
#

NetRange:       207.225.69.96 - 207.225.69.103
CIDR:           207.225.69.96/29
NetName:        Q0609-207-225-69-96
NetHandle:      NET-207-225-69-96-1
Parent:         CENTURYLINK-LEGACY-QWEST-INET-111 (NET-207-224-0-0-1)
NetType:        Reassigned
OriginAS:
Organization:   TESLA, INC. (TESLA-4)
RegDate:        2017-06-09
Updated:        2017-06-09
Ref:            https://rdap.arin.net/registry/ip/207.225.69.96


OrgName:        TESLA, INC.
OrgId:          TESLA-4
Address:        4330 SW MACADAM AVE
City:           PORTLAND
StateProv:      OR
PostalCode:     97239
Country:        US
RegDate:        2017-06-09
Updated:        2017-06-09
Ref:            https://rdap.arin.net/registry/entity/TESLA-4
```

Figure 29 — WHOIS confirmation: 207.225.69.96/29 belongs to TESLA, INC. (TESLA-4)



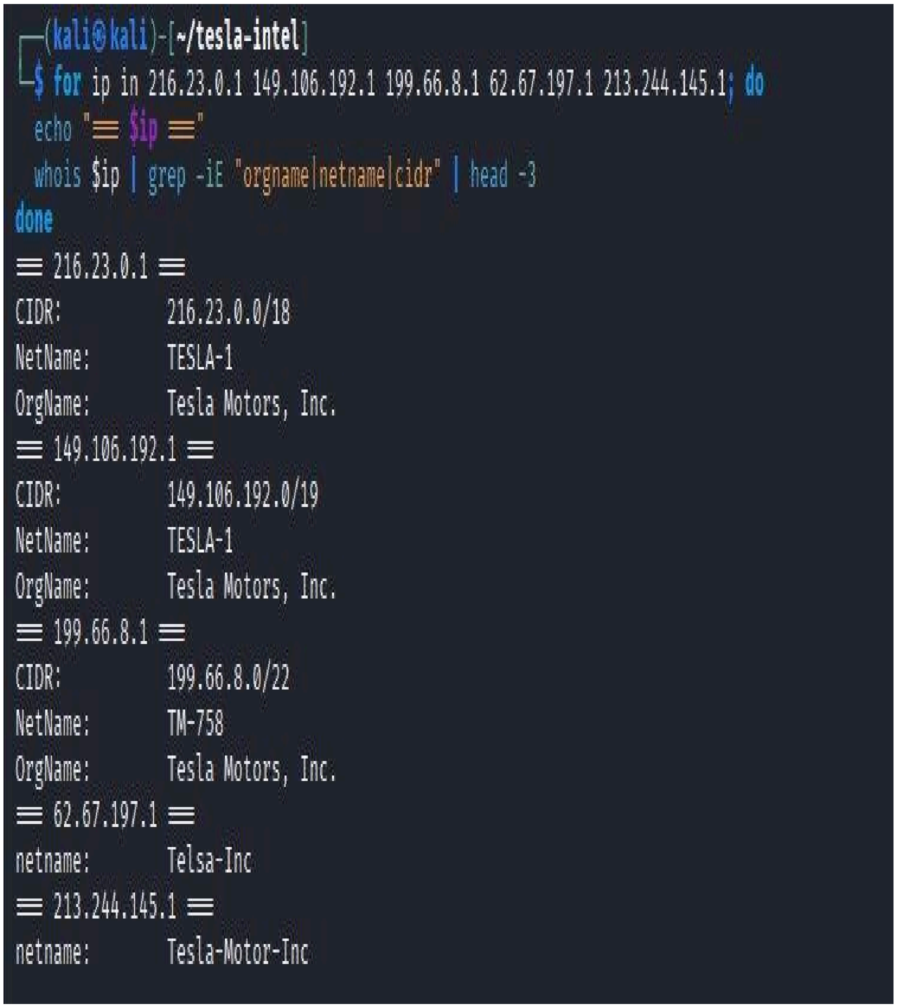
```
kali@kali: ~/tesla-intel
Session Actions Edit View Help

┌──(kali㉿kali)-[~/tesla-intel]
└─$ whois NET-63-145-195-80-1 2>/dev/null || whois 63.145.195.80 | grep -iE "orgname|netname|cidr|inetnum"

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#


#
# Query terms are ambiguous.  The query is assumed to be:
#     "n ! net-63-145-195-80-1"
#
# Use "?" to get help.
#

NetRange:       63.145.195.80 - 63.145.195.87
CIDR:           63.145.195.80/29
NetName:        Q0711-63-145-195-80
NetHandle:      NET-63-145-195-80-1
Parent:         CENTURYLINK-LEGACY-QWEST-INET-8 (NET-63-144-0-0-1)
NetType:        Reassigned
OriginAS:
Organization:   TESLA, INC. (TESLA-4)
RegDate:        2017-07-11
Updated:        2017-07-11
Ref:            https://rdap.arin.net/registry/ip/63.145.195.80


OrgName:        TESLA, INC.
OrgId:          TESLA-4
Address:        4330 SW MACADAM AVE
City:           PORTLAND
StateProv:      OR
PostalCode:     97239
Country:        US
RegDate:        2017-06-09
Updated:        2017-06-09
Ref:            https://rdap.arin.net/registry/entity/TESLA-4
```

Figure 30 — WHOIS confirmation: 63.145.195.80/29 belongs to TESLA, INC. (TESLA-4)

Full ASN Route Discovery via RADB

The RADB routing database was queried for all routes announced by AS394161 to discover the complete set of Tesla-owned IP prefixes:

Figure 31 — ARIN RDAP confirming AS394161 (TESLA) and AS394363 (TESLAENERGY)

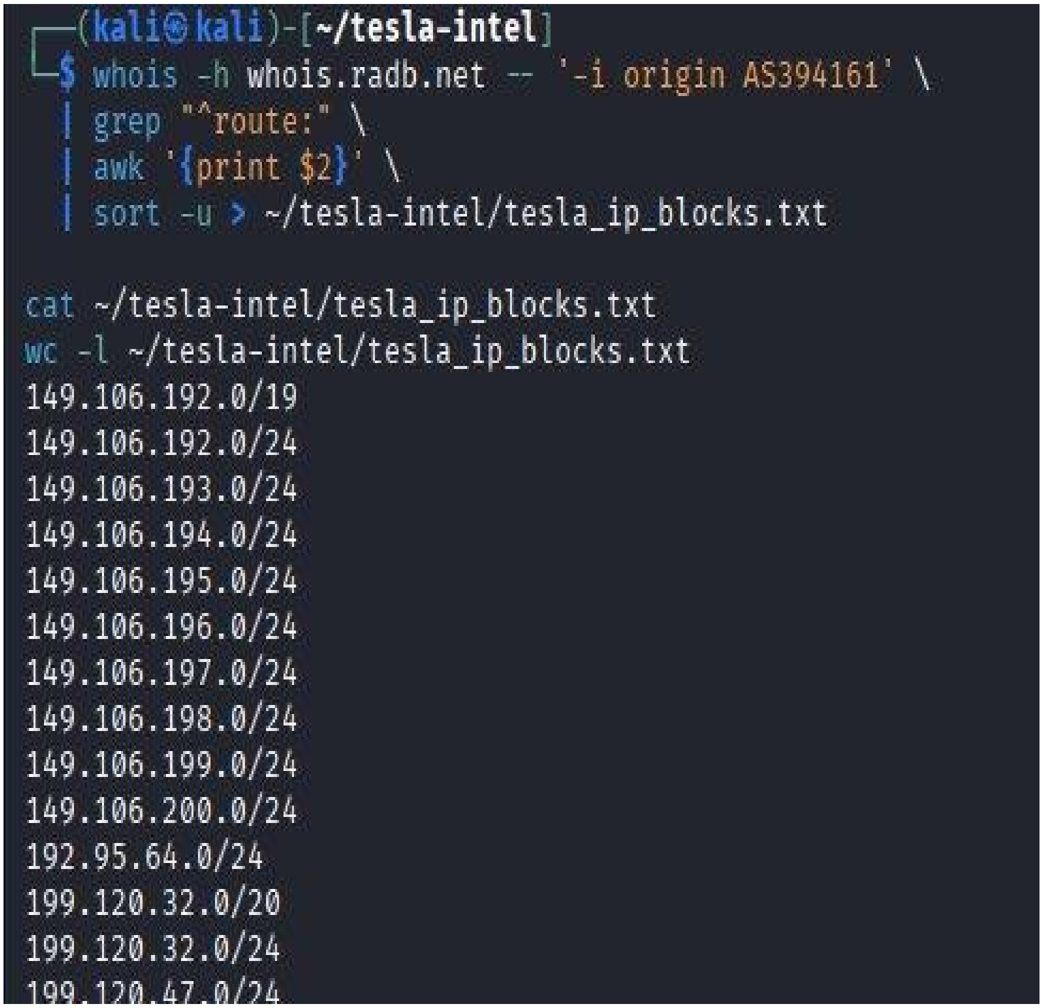


```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ for ip in 216.23.0.1 149.106.192.1 199.66.8.1 62.67.197.1 213.244.145.1; do
  echo "═══ $ip ═══"
  whois $ip | grep -iE "orgname|netname|cidr" | head -3
done
═══ 216.23.0.1 ═══
CIDR:           216.23.0.0/18
NetName:        TESLA-1
OrgName:        Tesla Motors, Inc.
═══ 149.106.192.1 ═══
CIDR:           149.106.192.0/19
NetName:        TESLA-1
OrgName:        Tesla Motors, Inc.
═══ 199.66.8.1 ═══
CIDR:           199.66.8.0/22
NetName:        TM-758
OrgName:        Tesla Motors, Inc.
═══ 62.67.197.1 ═══
netname:        Telsa-Inc
═══ 213.244.145.1 ═══
netname:        Tesla-Motor-Inc
```

Figure 32 — RADB routes for AS394161: full set of Tesla IP prefixes including 216.23.0.0/18, 149.106.192.0/19, 199.66.8.0/22

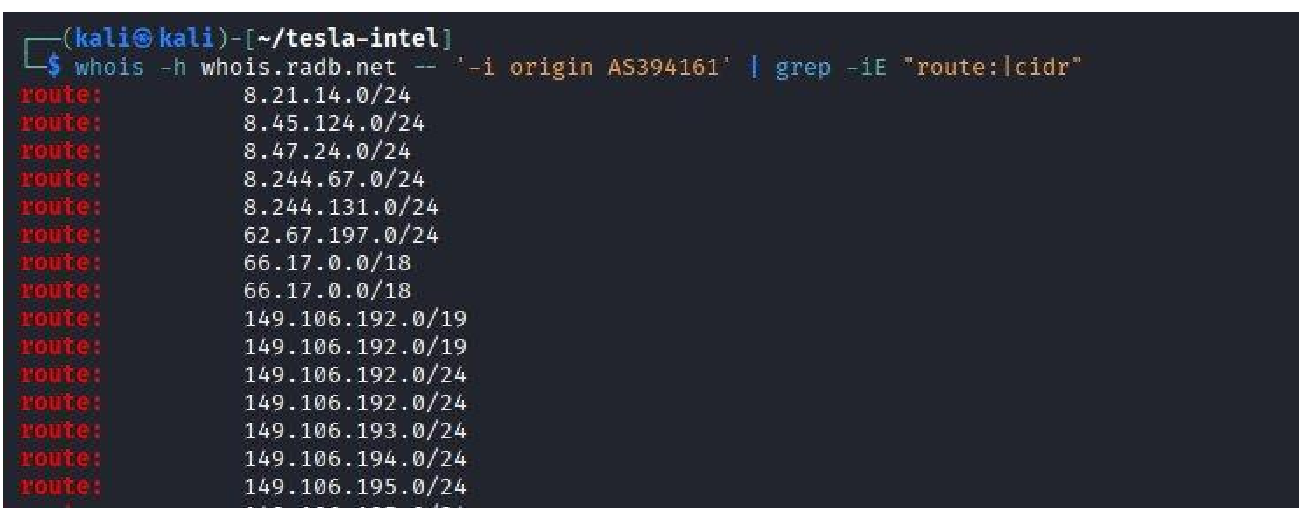
IP Block Verification

All newly discovered IP ranges were verified against WHOIS to confirm Tesla ownership:



```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ whois -h whois.radb.net -- '-i origin AS394161' \
  | grep "^route:" \
  | awk '{print $2}' \
  | sort -u > ~/tesla-intel/tesla_ip_blocks.txt

cat ~/tesla-intel/tesla_ip_blocks.txt
wc -l ~/tesla-intel/tesla_ip_blocks.txt
149.106.192.0/19
149.106.192.0/24
149.106.193.0/24
149.106.194.0/24
149.106.195.0/24
149.106.196.0/24
149.106.197.0/24
149.106.198.0/24
149.106.199.0/24
149.106.200.0/24
192.95.64.0/24
199.120.32.0/20
199.120.32.0/24
199.120.47.0/24
```

Figure 33 — Saved tesla_ip_blocks.txt with all confirmed Tesla-owned CIDR ranges



```
┌──(kali㉿kali)-[~/tesla-intel]
└─$ whois -h whois.radb.net -- '-i origin AS394161' | grep -iE "route:|cidr"
route:          8.21.14.0/24
route:          8.45.124.0/24
route:          8.47.24.0/24
route:          8.244.67.0/24
route:          8.244.131.0/24
route:          62.67.197.0/24
route:          66.17.0.0/18
route:          66.17.0.0/18
route:          149.106.192.0/19
route:          149.106.192.0/19
route:          149.106.192.0/24
route:          149.106.192.0/24
route:          149.106.193.0/24
route:          149.106.194.0/24
route:          149.106.195.0/24
```

Figure 34 — RADB WHOIS query listing IP prefixes announced by Tesla's Autonomous System (AS394161)

3.10 Phase 10 — Shodan Passive Intelligence

The Shodan search engine was queried with org:"Tesla Motors" to enumerate all publicly indexed Tesla IP addresses, open ports, and exposed services. This is a purely passive operation — Shodan's data is pre-collected.

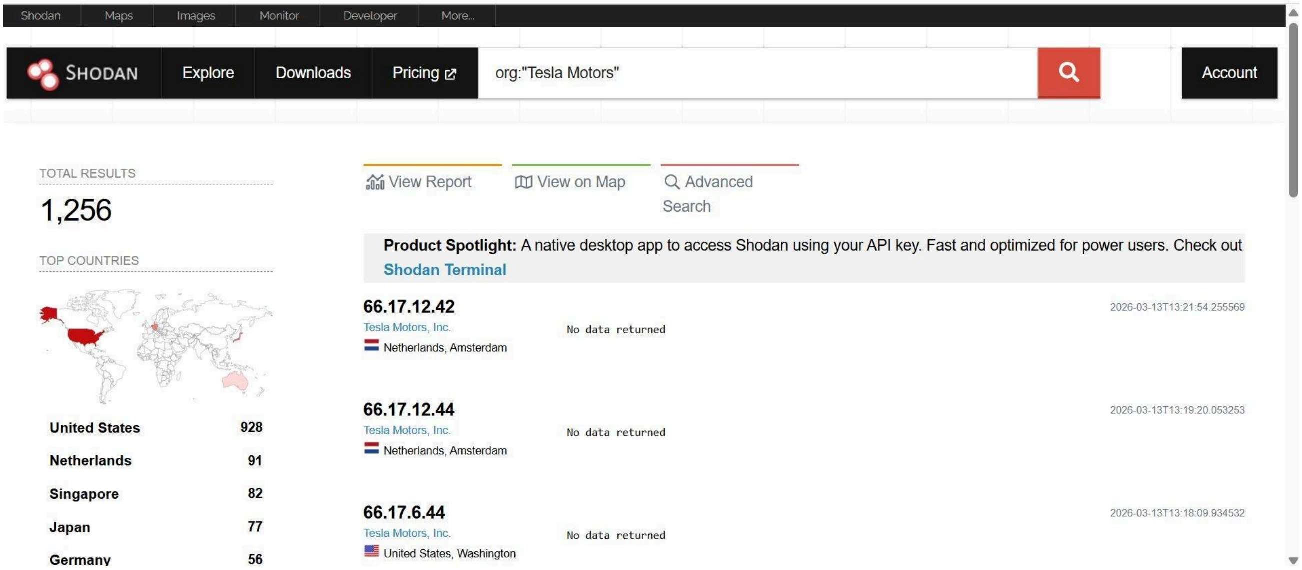


Figure 35 — Shodan search for org:"Tesla Motors"

Shodan results summary:

Category	Detail	Category	Detail
Total IPs indexed	1,256	Top Organization	Tesla Motors, Inc. (1,215)
United States	928	Netherlands	91 (Amsterdam hub)
Singapore	82 (APAC hub)	Japan	77 (Tokyo)
Germany	56 (Gigafactory Berlin)	Top Port	443 HTTPS (963 IPs)
Port 8883 (MQTT)	77 IPs — vehicle telemetry	Port 161 (SNMP)	35 IPs — network devices
Products: MQTT	14 instances	Products: ciscoSystems	14 instances
nginx	13 instances	F5 BigIP	8 instances

The presence of MQTT (port 8883) across 77 IPs is particularly notable — MQTT is the protocol used for vehicle-to-cloud telemetry communication, confirming Tesla's real-time vehicle data pipeline. The Cisco and F5 BigIP entries indicate enterprise-grade network infrastructure exposed on the internet perimeter.

4. Final Asset Inventory

The following tables summarize all confirmed Tesla assets discovered through this passive reconnaissance exercise.

4.1 IP Blocks (Confirmed Tesla-Owned)

CIDR Block	Size	NetName	Notes
66.17.0.0/18	/18 (16,384)	TESLA-1	Main cloud infrastructure
149.106.192.0/19	/19 (8,192)	TESLA-1	Vehicle network cloud
199.120.32.0/20	/20 (4,096)	TESLA-1	Gateway infrastructure
216.23.0.0/18	/18 (16,384)	TESLA-1	Large US block
199.66.8.0/22	/22 (1,024)	TM-758	Tesla Motors block
62.67.197.0/24	/24 (256)	Telsa-Inc	European range
213.244.145.0/24	/24 (256)	Tesla-Motor-Inc	European range
207.225.69.96/29	/29 (8)	TESLA-4	CenturyLink reassigned, Portland OR
63.145.195.80/29	/29 (8)	TESLA-4	CenturyLink reassigned, Portland OR

4.2 Autonomous System Numbers

ASN	Name	Description
AS394161	TESLA	Primary Tesla autonomous system
AS394363	TESLAENERGY	Tesla Energy division autonomous system

4.3 Subdomain Discovery Summary

Metric	Value
Total subdomains discovered	524
From crt.sh (SSL certificates)	505
From Assetfinder (OSINT)	53 (+19 unique)
Wildcard certificate entries	28
Live subdomains (httprobe)	205 (39% live rate)
HTTPS (live)	127

Metric	Value
HTTP → HTTPS redirects	78
Unique IP addresses (live)	70

4.4 Infrastructure & Technology Stack

Component	Technology	Evidence
CDN / DNS	Akamai + UltraDNS	NS records, IP WHOIS
Email	Microsoft Office 365	MX record: mail.protection.outlook.com
Web servers	nginx	Shodan service banners
Load balancers	F5 BigIP	Shodan service banners
Cloud providers	AWS + Azure	IP WHOIS (fleet-api, teslacmg*)
Container OS	AWS Bottlerocket	*.bottlerocket.tesla.com wildcard
Network gear	Cisco Systems	Shodan 14 instances
Vehicle protocol	MQTT (port 8883)	Shodan 77 IPs on port 8883
NTP	ntpd	Shodan 13 instances
VDI / Remote Access	*.vdi.tesla.com	Wildcard certificate
MDM / Device Mgmt	Microsoft SCCM (likely)	teslacmg* on Azure IPs

5. Conclusion

This exercise successfully mapped Tesla's complete public attack surface using exclusively passive, publicly available data sources. Starting from a single seed domain (tesla.com), the following assets were confirmed:

- 524 subdomains discovered, 205 confirmed live
- 9 Tesla-owned IP blocks totaling ~46,000+ IP addresses
- 2 Autonomous System Numbers (TESLA, TESLAENERGY)
- 1,256 IP addresses indexed by Shodan globally
- Complete technology stack identification (Akamai, AWS, Azure, nginx, F5, Cisco, MQTT)
- Global infrastructure presence confirmed across: US, Netherlands, Singapore, Japan, Germany

5.1 Key Takeaways

Tesla employs enterprise-grade security posture for its domain registration (MarkMonitor + EPP locks + privacy proxy) and CDN/DNS infrastructure (Akamai + UltraDNS), making traditional intelligence gathering techniques (Reverse WHOIS, Reverse NS) ineffective. However, Certificate Transparency logs — which cannot be blocked by any privacy measure — remain the most effective passive subdomain discovery technique against any HTTPS-enabled target.

The vn.cloud.tesla.com namespace and the extensive fleet-api, ownerapi, and signaling infrastructure visible through certificate logs provides a detailed picture of Tesla's vehicle-cloud architecture. The presence of MQTT on port 8883 across 77 Shodan-indexed IPs confirms the real-time vehicle telemetry pipeline.

5.2 Methodology Limitations

This assessment is bounded by the following limitations:

- No active scanning performed — Nmap, port scanning, and service probing were explicitly excluded
- No authentication or session data was collected from any live service
- Cloud provider IP ranges (AWS, Azure, Fastly) were not fully enumerated as they are shared infrastructure
- The TESLAENERGY ASN (AS394363) had no routes in RADB at time of query — may require further investigation
- Some crt.sh results may represent historical certificates for decommissioned services

5.3 Recommended Next Steps (Bug Bounty Context)

If this exercise were extended into an authorized bug bounty engagement (Tesla operates a program on Bugcrowd), the following would be the logical next steps:

- Technology fingerprinting on live endpoints using Wappalyzer / WhatCMS
- Login page enumeration using automated Python detection script (requests + BeautifulSoup)
- Historical exposure analysis using Wayback Machine for sensitive endpoints
- GitHub dorking for leaked credentials related to tesla.com
- Shodan deep-dive on MQTT endpoints (port 8883) for authentication configurations

End of Report

All activities were passive and legal. No systems were accessed or exploited.